CTM Workshop for Quarto

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Introduction

- First of all, thank you for inviting us to Lund.
- Today, we would like to present Quarto to you.
- (Most of you probably already know it.)
- For those who are new to Quarto, this presentation features a beginner's guide.

Table of contents

For Beginners

What is Quarto?

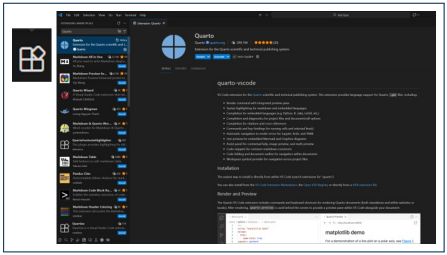
- Website: `https://quarto.org/`
- You write standard Markdown code exactly how you want it to appear.
- You can also embed code (Python, R, Julia, Observable, ...).
- During rendering, Quarto scans the file for code blocks. Each code block is processed by the appropriate engine, and the output is inserted into the document.
- Quarto then uses Pandoc to format the page using Bootstrap themes.
- Finally, it generates an HTML web page, or if you want other formats such as PDF, .docx, ...

How we use it

- We currently use it to demonstrate how Python can illustrate basic mathematics (like curve analysis) or help visualize matrix transformations.
- For this, we create website projects that include “pyodide-python”, a web-based Python environment allowing users to run code without local installation.
- Michael and others expanded a pyodide-python extension to include AI feedback. Michael can probably explain the technical details.
- Quarto can also be used for interactive learning sessions, including quizzes.

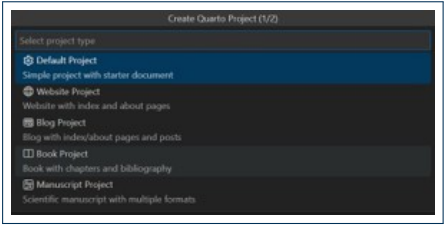
How to get started with Quarto

- Install an IDE of your choice (any text editor works).
- For this guide, we use Visual Studio Code.
- Install VSCode: `https://code.visualstudio.com/download`
- Install Quarto: `https://quarto.org/docs/get-started/`
- Then, install the Quarto extension for VSCode.
- Click the icon in the bottom left to find extensions.



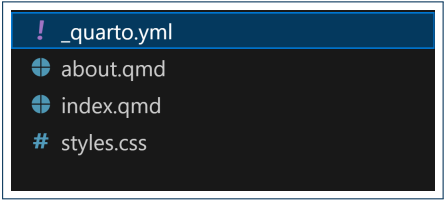
Getting to know Quarto (in VSCode)

- Go to File → New File
- When creating a new Quarto project, the following prompt should appear:
- Here, we select 'Website Project', as that fits our needs.



Getting to know Quarto (in VSCode)

- You should now have 4 files: `_quarto.yml`, `about.qmd`, `index.qmd`, `styles.css`
- These files form the foundation of a Quarto website. You can already render and preview the site (via the terminal with `quarto render` or the render button in the top right of any .qmd file).
- The `_quarto.yml` is the configuration file for the website; here you can adjust the core design and global settings.
- Each .qmd file has a YAML header (marked by `---` at the top and bottom). Here you can override settings for that specific page.



Quarto-Syntax

- `#, ##, ###` creates sections, subsections, subsubsections, etc.
- `:::{} ...:::` creates callout blocks or tabsets, depending on the content inside `{}`.
- `<details> </details>` creates a foldable HTML block. You can add a title using `<summary> </summary>` inside.
- ```` ```{} ``` ```` creates a code block. The content of `{}` defines the language.

Quarto Extensions

What are Quarto Extensions?

- Quarto extensions are community-contributed add-ons that enhance Quarto's functionality.
- They can add new **shortcodes**, **filters**, **formats**, or **journal templates**.
- Extensions are hosted on GitHub and managed directly through the Quarto CLI.
- The official listing of extensions can be found at:
  `https://quarto.org/docs/extensions/`

Installing and Using Extensions

Installing an extension:

- Run `quarto add user/repo` in your project directory.
- Example: `quarto add quarto-ext/fontawesome`
- The extension is stored locally in `_extensions/`.

Using an extension:

- **Filters:** add `filters: [extension-name]` to the YAML header.
- **Shortcodes:** use `{{< shortcode >}}` inline in your document.
- **Formats:** set `format: extension-name-html` in the YAML header.

Note

Extensions are project-local. You install them once per project and commit `_extensions/` to version control so collaborators don't need to reinstall.

Notable Quarto Extensions

Interactive & Code

- **pyodide** — run Python in the browser (WebAssembly)
- **webr** — run R in the browser (WebAssembly)
- **shinylive** — embed Shiny apps without a server

Learning & Teaching

- **quizdown** — multiple-choice and ordering quizzes
- **countdown** — in-slide countdown timer

Design & Layout

- **fontawesome** — {{< fa icon >}} shortcode for icons
- **lightbox** — click-to-zoom image galleries
- **roughnotation** — hand-drawn style annotations
- **animate** — CSS scroll animations

Publishing

- **academicons** — academic/research icons
- Journal templates (JASA, Elsevier, ...)

Pyodide-Python

Pyodide-Python

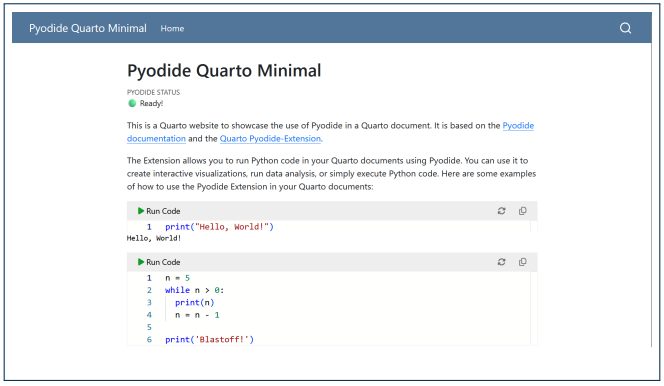
- Pyodide-Python is a web-based Python environment that allows you to run Python without local installation, as all computing is done in the browser.
- This is especially useful for giving students an easy entry into using Python for **Computational Thinking in Mathematics**.
- The extension can be installed via the terminal with `quarto add coatless-quarto/pyodide`.
- The extension is open-source and can be found at: `https://github.com/coatless-quarto/pyodide`

Pyodide-Python

- To use the extension in your .qmd files, you need to add `filters: - pyodide` to the YAML header.
- Afterward, you can use ```` ```{pyodide-python} ...``` ```` to embed Python code blocks into the page.
- For example:
  ```` ```{pyodide-python} print("Hello World") ``` ````
  will create a code block containing `print("Hello World")`, which can be edited and executed on the website.

```yaml
---
title: "Pyodide Quarto Minimal"
filters:
  - pyodide
---
```
```

Pyodide-Python in Action



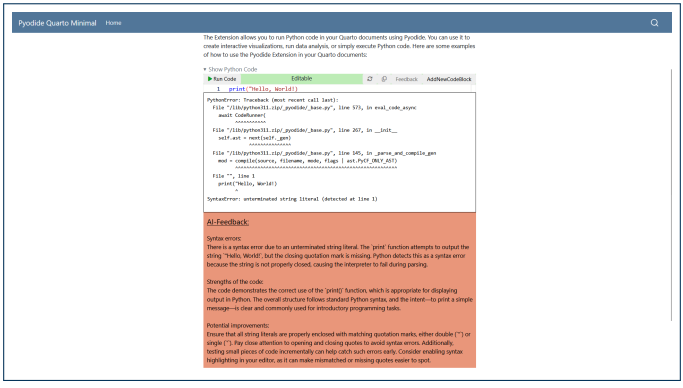
A minimal example of a website using the Pyodide-Extension can be found in this repository or directly at: `https://gitlab.ruhr-uni-bochum.de/ctm/lund-meeting/-/tree/main/Pyodide%20Quarto%20Minimal?ref_type=heads`

AI-Feedback within Pyodide Python

Pyodide-Python with AI-Feedback

- At Ruhr University Bochum, we modified the **coatless-pyodide** extension to include AI feedback.
- The extension works the same as the original but adds the ability to ask an AI for help with the code.
- To use this feature, users must provide a valid **API Key** and **Base URL**.

Pyodide-Python with AI-Feedback in Action



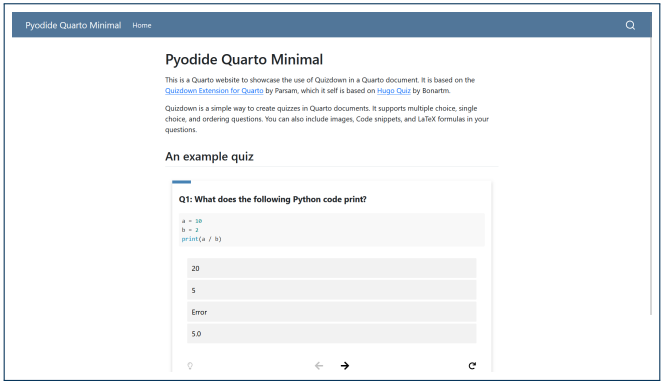
A minimal example of a website using the modified Pyodide-Extension can be found in this repository or directly at: `https://gitlab.ruhr-uni-bochum.de/ctm/lund-meeting/-/tree/main/Pyodide-Feedback%20Minimal?ref_type=heads`

Quarto Quizzes

Quizdown

- Gamification can greatly enhance learning, so we wanted to provide a gamified experience in our tutorials.
- For this purpose, we looked and found the **Quizdown** extension: `https://github.com/parmsam/quarto-quizdown`
- Quizdown allows you to create single-choice, multiple-choice, and ordering quizzes without much extra effort.

Quizdown in Action



A minimal example of a website with Quizdown can be found in this repository or directly at: `https://gitlab.ruhr-uni-bochum.de/ctm/lund-meeting/-/tree/main/Quizdown%20Minimal?ref_type=heads`

Shinylive

Shinylive

- In further enhancing the interactive experience we looked for something that would allow us to give users direct feedback on mathematical concepts.
- For that we looked into Shiny, specifically the shinylive extension for Quarto: `https://github.com/quarto-ext/shinylive`
- Shinylive is an extension that lets you embed Shiny apps into your Quarto website.
- The app runs entirely in the browser (via WebAssembly), and the code can be edited live with sliders, making the consequences immediately visible.

Shinylive in Action

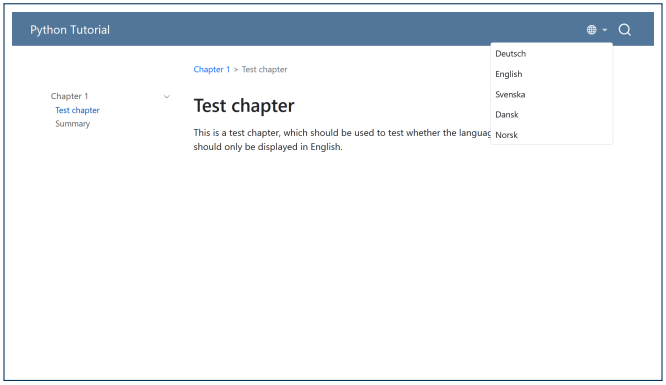
A minimal example of a website using the shinylive Extension can be found in this repository or directly at: `https://gitlab.ruhr-uni-bochum.de/ctm/lund-meeting/-/tree/main/Shinylive%20Minimal?ref_type=heads`

Multilanguage Website Projects

Multilanguage Website Projects

- As this is an international project, we wanted to give users the ability to switch between languages in Quarto.
- Currently, there is no official feature to switch languages within a specific part of the website; it only supports redirecting to a different homepage.
- Therefore, we wrote a small Python script that modifies the navbar to include a dropdown menu, allowing users to switch languages while staying on the same page.
- Quarto's `preview` command does not support the switching between languages, so you won't be able to properly preview your page using `quarto preview`
- Instead, run the included Python script to host the site locally and preview it in the browser.

Multilanguage Website Projects in Action



A minimal example of a website with multi-language support (I hope the AI translations aren't too bad!) can be found in this repository or directly at: `https://gitlab.ruhr-uni-bochum.de/ctm/lund-meeting/-/tree/main/Multilanguage%20Website%20Setup?ref_type=heads`

Conclusion

Conclusion

- Quarto is a powerful tool for creating interactive and engaging educational content, especially in the context of Computational Thinking in Mathematics.
- There is a vast number of extensions available, providing many possibilities to enhance the student experience.
- As some of you have already seen, Quarto is a user-friendly tool for creating scripts, as it allows you to combine code and output in a single file.

Resources

All materials from this presentation and the slides can be found at:

`https://gitlab.ruhr-uni-bochum.de/ctm/lund-meeting`